Exact Product-Kernel Gelation and the Smoluchowski–Flory Postgel Boundary

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Abstract

We give a convention-complete solution atlas for monodisperse discrete coagulation with multiplicative kernel. Cayley tree coefficients close every cluster concentration before gelation, while rooted and unrooted tree functions give the mass and number generating functions. The second moment diverges at $t = 1$, where the concentrations have the critical $k^{-5/2}$ tail. We then resolve a postgel ambiguity explicitly. With the Smoluchowski finite-sol loss, one Stockmayer continuation freezes the critical shape and has sol mass $1/t$. With gel-reactive Flory loss, the pregel coefficient formula continues and has sol mass $q = -W_0(-te^{-t})/t$. The two families agree at $t = 1$ but direct substitution shows that they solve different equations afterward. No universal weak-solution uniqueness is claimed. Exact independent reconstruction, symbolic identities, canonical replay, and hostile mutations audit the theorem.

1 Frozen kinetic convention

For $k \geq 1$, set

$$\dot{c}_k = \frac{1}{2}\sum_{i+j=k} ijc_ic_j - kc_kM_1(t), \qquad M_q(t) = \sum_{k\geq 1} k^q c_k(t), \tag{1}$$

with $c_1(0) = 1$ and $c_k(0) = 0$ for $k > 1$. Thus $K(i, j) = ij$, the unordered gain has factor $1/2$, and (1) uses only finite-cluster mass in its loss. A different gel-reactive loss will be introduced only after its convention is named.

Define the rooted and unrooted labelled-tree series

$$T(u) = \sum_{k\geq 1} \frac{k^{k-1}}{k!}u^k, \qquad U(u) = \sum_{k\geq 1} \frac{k^{k-2}}{k!}u^k. \tag{2}$$

Cayley's formula and Lagrange inversion give $T = ue^T$, $U = T - T^2/2$, and $uU'(u) = T(u)$.

Theorem 1 (Exact pregel solution and gel point)**.** *For $0 \leq t \leq 1$,*

$$c_k(t) = a_kt^{k-1}e^{-kt}, \qquad a_k = \frac{k^{k-2}}{k!}. \tag{3}$$

If $u = tze^{-t}$, the mass and number generating functions are

$$G(z, t) = \sum_{k\geq 1} kc_kz^k = \frac{T(u)}{t}, \qquad C(z, t) = \sum_{k\geq 1} c_kz^k = \frac{U(u)}{t}, \tag{4}$$

as formal power series in z, and analytically on the principal convergence disk (in particular $0 \leq z \leq 1$), with continuous $t = 0$ values. For $t < 1$,

$$M_0 = 1 - \frac{t}{2}, \quad M_1 = 1, \quad M_2 = \frac{1}{1-t}, \quad M_3 = \frac{1}{(1-t)^3}. \tag{5}$$

At $t_g = 1$,

$$c_k(1) \sim \frac{1}{\sqrt{2\pi}} k^{-5/2}, \tag{6}$$

so M_2 diverges. For fixed $0 < t < 1$,

$$c_k(t) \sim \frac{k^{-5/2}}{\sqrt{2\pi}\, t} [te^{1-t}]^k. \tag{7}$$

Proof. Coefficient extraction from $T = ue^T$, or the standard labelled-tree edge decomposition, gives

$$(k-1)a_k = \frac{1}{2} \sum_{i+j=k} ij a_i a_j. \tag{8}$$

At $t = 0$, equation (3) means its continuous limit: $c_1 = 1$ and $c_k = 0$ for $k > 1$, so no 0^0 convention is left implicit. Differentiating (3), using (8), and imposing $M_1 = 1$ proves (1) coefficient by coefficient. Substitution in (2) gives (4). On the branch reached from zero, $T(te^{-t}) = t$ for $t \le 1$, so $U = t - t^2/2$. Applying $z\partial_z$ to G and using $uT'(u) = T/(1-T)$ gives (5); equivalently $\dot{M}_2 = M_2^2$ and $\dot{M}_3 = 3M_2M_3$. Stirling's formula applied to (3) gives (6) and (7), and (6) makes $\sum k^2 c_k$ divergent. □

2 Two inequivalent postgel closures

Proposition 1 (Finite-sol Smoluchowski/Stockmayer branch)**.** *For equation (1), one explicit continuation for $t \ge 1$ is*

$$c_k^{\mathrm{S}}(t) = a_k \frac{e^{-k}}{t}. \tag{9}$$

It is continuous coefficientwise at $t = 1$ and has

$$M_0^{\mathrm{S}} = \frac{1}{2t}, \qquad M_1^{\mathrm{S}} = \frac{1}{t}, \qquad M_2^{\mathrm{S}} = \infty. \tag{10}$$

Proof. At the critical argument, $T(e^{-1}) = 1$ and $U(e^{-1}) = 1/2$, giving (10). By (8), the gain in (1) is $(k-1)c_k^{\mathrm{S}}/t$; the loss is $kc_k^{\mathrm{S}} M_1^{\mathrm{S}} = kc_k^{\mathrm{S}}/t$. Their difference is $-c_k^{\mathrm{S}}/t = \dot{c}_k^{\mathrm{S}}$. The critical tail (6), merely scaled by $1/t$, makes M_2 infinite. □

Now change the postgel equation: keep the same gain but replace its loss by $-kc_kM_1(0) = -kc_k$. This is the gel-reactive Flory convention; finite clusters may continue to collide with the gel.

Proposition 2 (Gel-reactive Flory branch)**.** *For the Flory loss and $t \ge 1$,*

$$c_k^{\mathrm{F}}(t) = a_k t^{k-1} e^{-kt}. \tag{11}$$

For $t > 1$, let

$$r = -W_0(-te^{-t}) \in (0,1), \qquad q = r/t. \tag{12}$$

Then $q = e^{-t(1-q)}$ and

$$M_0^{\mathrm{F}} = q - \frac{t}{2}q^2, \quad M_1^{\mathrm{F}} = q, \quad M_2^{\mathrm{F}} = \frac{q}{1-r}, \quad M_3^{\mathrm{F}} = \frac{q}{(1-r)^3}. \tag{13}$$

Proof. Equation (8) gives gain $(k-1)c_k^{\mathrm{F}}/t$; Flory loss is kc_k^{F}. Their difference equals $[(k-1)/t - k]c_k^{\mathrm{F}} = \dot{c}_k^{\mathrm{F}}$. The principal Lambert branch solves $re^{-r} = te^{-t}$ and selects $r < 1$, yielding the fixed-point equation for q. Equations (4) and $U = T - T^2/2$, evaluated at $T = r$, give M_0 and M_1. Two applications of $z\partial_z$ give the remaining moments. □

The same fixed-point equation also has the formal root $q = 1$: for $t > 1$ it comes from $W_{-1}(-te^{-t}) = -t$. The coefficient generating series selects the principal branch W_0, hence $r < 1$ and $q < 1$; the W_{-1} root is not the finite-cluster mass of (11).

Equations (9) and (11) agree at $t = 1$. For every $t > 1$, however, their loss masses are respectively $1/t$ and 1, and their sol masses are respectively $1/t$ and q. Agreement at one boundary therefore cannot identify the two dynamical closures. We verify these explicit continuations but make no claim of uniqueness among all weak postgel solutions.

3 Source and collision boundary

McLeod [1, 2] is a primary owner for the infinite nonlinear system and its classical exact analysis. Ziff–Stell [3] treat polymer gelation conventions; Normand–Zambotti [4] and Norris [5] delimit global solution and uniqueness questions. We claim no priority for the tree law, gel exponent, or named postgel branches. The local contribution is an executable convention firewall.

No earlier local paper owns this kinetic system. Linear branching and the finite Kingman coalescent have different states and generators; neither has the concentration-mass gel boundary above.

References

[1] J. B. McLeod, *On an Infinite Set of Non-Linear Differential Equations*, Quarterly Journal of Mathematics 13 (1962), 119–128. DOI: 10.1093/qmath/13.1.119.

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[3] R. M. Ziff and G. Stell, *Kinetics of polymer gelation*, Journal of Chemical Physics 73(7) (1980), 3492–3499. DOI: 10.1063/1.440502.

[4] C. Normand and L. Zambotti, *Uniqueness of post-gelation solutions of a class of coagulation equations*, Annales de l'Institut Henri Poincaré C 28(2) (2011), 189–215. DOI: 10.1016/j.anihpc.2010.10.005.

[5] J. R. Norris, *Smoluchowski's coagulation equation: uniqueness, nonuniqueness and a hydrodynamic limit for the stochastic coalescent*, Annals of Applied Probability 9(1) (1999), 78–109. DOI: 10.1214/aoap/1029962598.

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; no target arithmetic, target-zero, universal postgel-uniqueness, or Hilbert–Pólya claim. **Data and code.** The exact coefficient ledger and all audits accompany HCS-C228. **AI-use disclosure.** Generative tools assisted drafting and code generation; the released internal chain checks the displayed formulas and metadata. This is not external peer review.